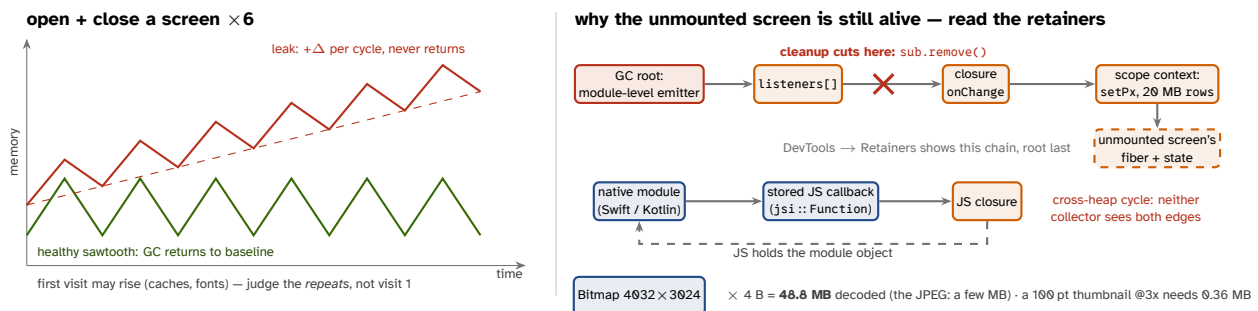


In one line: A leak is **unwanted reachability** from a **GC root** after the screen that needed it is gone. RN has **two memories**: the **Hermes JS heap** (DevTools heap snapshots) and **native memory** (views, bitmaps, modules — Xcode / Android Studio only). Proof is a **staircase**: the same growth on every repeat of an action.



How it works

- **JS heap:** Hermes' tracing GC frees anything unreachable — cycles included. A JS leak is always a *path from a root*: module-level variables, globals, native event registries, timers, pending promises.
- **Native memory:** views, **decoded bitmaps** (Fresco / SDWebImage / Glide caches), WebViews, module state — ARC on iOS, ART GC + native heap on Android. Invisible to a JS heap snapshot.
- **Across the boundary:** native holding a JS callback or JS holding a HostObject keeps the other side alive; a cycle through both heaps is not collected.
- **Mounted $\neq$ leaked, but costs:** stack screens under the top and visited tabs stay mounted — replace/reset finished flows.

Usual causes

- Listeners/subscriptions never removed (AppState, Keyboard, Linking, NativeEventEmitter, store/WebSocket subscriptions).
- setInterval/setTimeout/requestAnimationFrame not cleared; fetches resolving into unmounted components.
- Closures capturing big objects (one scope = one shared context); module-level caches with no bound (Map of responses, images, blobs).
- Native: a module keeping callbacks/promises, a strong delegate/block cycle, an Android Activity/ReactContext stored in a static field.

Example — leaking effect, fixed

```
function Ticker({ symbol }: { symbol: string }) {
  const [px, setPx] = useState<number>();
  useEffect(() => {
    const sub = AppState.addEventListener('change', onChange);
    const id = setInterval(poll, 1000);
    const ctrl = new AbortController();
    fetch(urlFor(symbol), { signal: ctrl.signal })
      .then(r => r.json())
      .then(setPx)
      .catch(e => { if (e.name !== 'AbortError') report(e); });
    return () => { // FIX: undo everything above
      sub.remove(); clearInterval(id); ctrl.abort();
    };
  }, [symbol]); // re-runs: cleanup of the OLD symbol first
}
const cache = new Map<string, Row[]>(); // grows forever -> LRU(50)
```

Hunting a JS leak — 3 snapshots

1. DevTools **Memory**. Do the action once (warm caches), return: snapshot **A**.
2. Repeat **N** times (e.g. 5): snapshot **B**; once more: **C**.
3. **Comparison** view B vs A, sort by # Delta / Size Delta: suspects grow by a multiple of N and are still there in C.
4. Select one instance → **Retainers**, read to the root; the first edge *your* code owns is the bug. **Retained size** = what freeing it gives back.

Native side + real footprint

- **iOS:** Xcode *Debug Memory Graph* (purple “!” = leaked cycles; Malloc Stack Logging adds backtraces); Instruments *Allocations* (Mark Generation per repeat) + *Leaks*.
- **Android:** Studio *Heap Dump* — shallow / retained / native size, depth, *Show activity/fragment leaks*; *Track Memory Consumption* (Java/Kotlin or native); LeakCanary (a Profiler task since Studio Panda). Totals: adb shell dumpsys meminfo <pkg> (Java/Native heap, Graphics, TOTAL PSS). Memory + bitmap memory are Play vitals.
- Heap flat but RSS climbing ⇒ it is native: images first.

Images — the maths and the fixes

- Decoded = **width** × **height** × **4 bytes** (RGBA8), whatever the file size. Request *display size* × *scale* from the CDN.
- Android <Image resizeMode="resize"> downsamples *before* decode (for images much larger than the view). Image.prefetch fills the *disk* cache.
- **FlashList** recycles a fixed cell pool → live bitmaps stay bounded.

Traps

- “No warning, so no leak” — React 18 *removed* the “state update on an unmounted component” warning; cleanup is still your job.
- freezeOnBlur/lazy tabs reduce *renders*, not memory held.

Remember

Repeat · snapshot · compare · retainers. JS heap flat ⇒ bitmaps.